Routing Protocols and Routing Attacks

IE3122: Network Security | Week 5

How routers trust each other, and what happens when that trust is abused

Today's Agenda

- Quick overview of routing protocols: RIP, OSPF, EIGRP, BGP
- Why routing protocols are attackable
- Five specific routing attacks
- Defenses: authentication, passive-interface, filtering, RPKI
- Bringing it together: layered defense for routing

Learning Outcomes for Today

LO1

- Describe common network vulnerabilities and attacks, today specifically at the routing layer.

LO2

- Evaluate different mitigation techniques for vulnerabilities and make recommendations.

- *By the end of this session you should be able to explain how a routing protocol decides what to trust, how that trust can be abused, and which defense actually closes the gap.*

Routing Protocol Fundamentals

How routers learn to trust each other

Why Routing Protocols Matter

- Recall Week 1: a router forwards packets based on its routing table. Routing protocols exist so routers can build that table automatically, by exchanging information with their neighbors instead of relying on manual configuration alone.
- This automation is powerful at scale. Large networks and the Internet itself simply couldn't run on manually maintained routes.
- But it means every router is placing real trust in whatever its neighbors tell it. Everything in this lecture follows from asking: what happens when that trust is misplaced?

RIP: Routing Information Protocol

- Type: distance-vector. Each router only knows what its direct neighbors tell it, and picks the path with the fewest hops.
- Simple to configure, but scales poorly and converges slowly on larger networks.
- No authentication in RIPv1 at all; RIPv2 added optional plaintext or MD5 authentication, but it's frequently left disabled in practice.

RIP: Metric and Messages

- Metric: hop count only. Every link counts as 1 hop regardless of its actual bandwidth or reliability, capped at a maximum of 15. A route needing 16 hops is treated as unreachable, which is part of why RIP struggles on larger networks.
- Messages: RIP Request (a router asking for routing information) and RIP Response (the reply, also sent unsolicited every 30 seconds by default to advertise the full routing table).
- Transport: runs directly over UDP, port 520, using broadcast in RIPv1 and multicast (224.0.0.9) in RIPv2.
- Security angle: because full routing table updates go out every 30 seconds to anyone listening on the segment, an attacker doesn't need to do anything sophisticated to see the whole picture, or to inject a convincing-looking Response of their own.

OSPF: Open Shortest Path First

- Type: link-state. Every router builds a full map of the network topology (within its area) and calculates the best path itself, rather than trusting a neighbor's summary.
- Organised into areas for scalability, with Area 0 (the backbone) connecting all others.
- Supports MD5 authentication between neighbors. This is one of the two protocols you'll actually configure and harden in this week's lab.

OSPF: Metric and Messages

- Metric: cost, calculated from interface bandwidth (a reference bandwidth divided by the interface's actual bandwidth). Faster links get a lower cost, so OSPF's path selection reflects real link speed, unlike RIP's flat hop count.
- Messages: Hello (discovers and maintains neighbor relationships), Database Description or DBD (summarises the link-state database during initial sync), Link State Request or LSR (asks for missing details), Link State Update or LSU (carries the actual Link State Advertisements), and Link State Acknowledgment or LSAck.
- Transport: runs directly over IP, protocol number 89 (not over UDP or TCP), using multicast addresses 224.0.0.5 (all OSPF routers) and 224.0.0.6 (designated routers).
- Security angle: because Hello packets are how OSPF forms a neighbor relationship in the first place, that's exactly the message MD5 authentication has to protect before any topology information is ever exchanged.

EIGRP: Enhanced Interior Gateway Routing Protocol

- Type: advanced distance-vector (Cisco calls it ‘hybrid’). Uses the DUAL algorithm to converge faster than classic distance-vector protocols like RIP, while staying simpler to configure than OSPF.
- Metric factors in bandwidth and delay, not just hop count, giving more realistic path selection.
- Also supports MD5 authentication between neighbors: the second protocol you'll configure and harden this week.

EIGRP: Metric and Messages

- Metric: a composite value combining bandwidth and delay by default (reliability and load are also part of the formula but disabled by default). This gives EIGRP a more realistic picture of path quality than a simple hop count.
- Messages: Hello (maintains neighbor adjacency, similar role to OSPF's), Update (carries actual routing information), Query and Reply (used by the DUAL algorithm to find a new loop-free path when a route fails), and Acknowledgment.
- Transport: runs directly over IP, protocol number 88, using multicast address 224.0.0.10.
- Security angle: DUAL's Query/Reply exchange is exactly what an attacker's false advertisements interact with during Routing Table Poisoning, since a convincing false route can trigger the same recalculation process a genuine link failure would.

BGP: Border Gateway Protocol

- Type: path-vector. The only exterior gateway protocol in this list, used to route between Autonomous Systems (AS), not within one. This is the protocol that holds the entire Internet together.
- A router advertises which prefixes (address blocks) it can reach and via which AS path; neighbors largely take that advertisement on trust.
- Supports MD5 authentication (a password per neighbor session), but Packet Tracer's BGP support is limited enough that we won't rely on configuring it hands-on this week.

BGP: Metric and Messages

- Metric: BGP doesn't use a single simple metric the way the other three do. Path selection runs through an ordered list of path attributes (weight, local preference, AS-path length, origin, and others), with AS-path length being the simplest to reason about: a shorter path through fewer Autonomous Systems is generally preferred.
- Messages: OPEN (establishes a session with a neighbor), UPDATE (advertises new routes or withdraws ones no longer valid, exactly the message type involved in a prefix hijack), KEEPALIVE (confirms the session is still alive), and NOTIFICATION (reports an error and closes the session).
- Transport: unlike the other three protocols, BGP runs over TCP, port 179, rather than directly over IP or UDP. This is also why BGP's MD5 authentication is technically a TCP-level protection (an MD5 signature carried in the TCP header), not a routing-protocol-level one.
- Security angle: the UPDATE message is the entire attack surface for prefix hijacking. Whoever can get an UPDATE accepted, truthfully or not, controls where that traffic goes next.

The Four Protocols at a Glance

Protocol	Type	Metric	Transport	Authentication
RIP	Distance-vector	Hop count (max 15)	UDP 520	Optional, rarely enabled
OSPF	Link-state	Cost (bandwidth-based)	IP protocol 89	MD5, well supported
EIGRP	Advanced distance-vector	Bandwidth + delay composite	IP protocol 88	MD5, well supported
BGP	Path-vector	AS-path length + attributes	TCP 179	MD5 possible, limited adoption

Notice BGP is the only one running over TCP rather than directly over IP or UDP, which is exactly why its MD5 authentication works differently to the others.

Why Routing Trust Is Exploitable

- Same theme as every week so far: by default, none of these protocols verify that a neighbor advertising routes is actually authorised to do so.
- A router that accepts a routing update has no built-in way to check whether the advertised route is truthful, current, or even physically plausible.
- The consequences scale with the protocol's reach: a bad RIP route affects one small network, while a bad BGP announcement can affect the global Internet, as today's real-world example will show.

Routing Attacks

Five ways trust in a routing protocol can be abused

Five Attacks at a Glance

Attack	What it targets
Routing Table Poisoning	Injecting false routes into a distance-vector protocol
Unauthorised Route Injection	A rogue router joining the routing domain uninvited
Route/Prefix Hijacking	Advertising ownership of address space that isn't yours (BGP)
Blackhole / Sinkhole Attack	Redirecting traffic to a dead end or attacker-controlled device
Routing DoS / Instability Attack	Flooding updates to force constant recalculation

Routing Table Poisoning: Overview & Technique

- Overview: the attacker injects false route advertisements into a distance-vector protocol, convincing routers to update their tables with incorrect paths.
- Vulnerability exploited: distance-vector protocols like RIP propagate route information hop-by-hop based on what a neighbor claims, with no independent way to verify the claim.
- Technique: a compromised or rogue device connected to the network advertises a very attractive-looking (low hop count) but false route to a target network, and legitimate routers propagate it further.
- Impact: traffic intended for the real destination gets misrouted, enabling denial of service, interception, or both, depending on where the false route actually points.

Routing Table Poisoning: Mitigation

- Route authentication (MD5) on neighbor relationships stops an unauthenticated device from having its advertisements accepted at all.
- Route filtering / distribute-lists: explicitly define which routes are acceptable to learn or advertise on a given interface, rejecting anything implausible.
- Passive-interface on any interface that doesn't need to participate in the routing protocol at all removes the attack surface outright on those links.

ATTACK 2

Unauthorised Route Injection: Overview & Technique

- Overview: an attacker connects a rogue router (or a device impersonating one) directly into the routing domain and forms a full neighbor relationship, not just a single false advertisement.
- Vulnerability exploited: without authentication, any device speaking the right routing protocol on the right network segment can become a fully trusted neighbor.
- Technique: the rogue router advertises itself as a legitimate participant, then either injects false routes continuously or simply gains visibility into the entire routing domain's topology.
- Impact: broader and more persistent than a single poisoning attempt: the attacker has an ongoing foothold, not a one-time injection.

Unauthorised Route Injection: Mitigation

- Neighbor authentication (MD5) is the direct fix: a rogue device without the correct key simply cannot form a neighbor relationship in the first place.
- Passive-interface on access-layer and other non-router-facing interfaces prevents a rogue device plugged into the wrong port from ever seeing routing traffic at all.
- Physical port security (recall Week 3) is a genuine complementary control here. If the rogue device can't reach a routing-capable interface physically, it can't attempt this attack regardless of protocol-level defenses.

Route/Prefix Hijacking: Overview & Technique

- Overview: an AS advertises a route for an IP prefix it doesn't actually own or isn't authorised to originate. A BGP-specific attack, given BGP is what governs address space ownership across the Internet.
- Vulnerability exploited: BGP has no built-in mechanism to verify that an AS advertising a prefix is its legitimate owner. Neighbors, and their neighbors in turn, largely accept advertisements on trust.
- Technique: the attacker (or a misconfigured legitimate network) announces a route for someone else's address block, often a more specific prefix, since BGP prefers the most specific match available.
- Impact: traffic meant for the real owner gets redirected globally, not just locally, exactly what happened in a well-documented real-world incident, next.

Real-World Example: The 2008 YouTube Hijack

- On 24 February 2008, Pakistan Telecom (AS17557) was ordered by its government to block YouTube domestically, and attempted this by advertising a route for part of YouTube's address block (208.65.153.0/24, a more specific slice of YouTube's actual 208.65.152.0/22).
- This was only ever meant to be visible inside Pakistan, but Pakistan Telecom's upstream provider, PCCW (AS3491), did not validate the advertisement and propagated it to the rest of the Internet.
- Because BGP prefers more specific routes, most of the global Internet started sending YouTube-bound traffic to Pakistan Telecom instead, effectively taking YouTube offline worldwide for roughly two hours, until PCCW disconnected Pakistan Telecom.
- This wasn't even a deliberate global attack. It was a local censorship attempt with no malicious intent to hijack anything outside Pakistan. That's exactly what makes it such a powerful illustration of how little verification BGP does by default.

Route/Prefix Hijacking: Mitigation

- RPKI (Resource Public Key Infrastructure) lets an address block's legitimate owner cryptographically sign a Route Origin Authorisation (ROA), so other networks can verify an advertisement before accepting it. This is the real-world fix that emerged after incidents like the YouTube hijack.
- Route filtering at provider boundaries: an upstream provider should only accept the specific prefixes it has agreed a customer is authorised to announce, exactly the validation PCCW skipped in 2008.
- This is theory-only for our purposes: RPKI and provider-level BGP filtering aren't things we can configure in Packet Tracer, but understanding why they exist is directly relevant to LO2's evaluation skill.

Blackhole / Sinkhole Attack: Overview & Technique

- Overview: the attacker manipulates routing so that traffic for a target is directed to a dead end (blackholed: silently dropped) or to a device they control (sinkholed: captured for analysis or further attack).
- Vulnerability exploited: the same lack of route verification as the other attacks in this section. The difference here is entirely in what the attacker does with the misdirected traffic, not in how they achieve the misdirection.
- Technique: often achieved via the same mechanisms as poisoning or unauthorised injection, just aimed specifically at creating a non-existent or attacker-controlled next hop for a target network.
- Impact: a blackhole is a denial of service; a sinkhole additionally lets the attacker inspect (and potentially tamper with) traffic before optionally forwarding it on, which can make the attack harder to notice.

Blackhole / Sinkhole Attack: Mitigation

- The same core defenses apply as for the attacks that enable it: route authentication, route filtering, and passive-interface all reduce the attacker's ability to influence next-hop selection in the first place.
- Monitoring for unexpected traffic drops or unusual latency/path changes is a useful detective control: a legitimate destination silently becoming unreachable is a strong signal worth alerting on.
- This attack is a good reminder that a single root cause (unverified routing trust) can be aimed at different outcomes, echoing Week 3's multi-function attack discussion.

Routing DoS / Instability Attack: Overview & Technique

- Overview: rather than trying to redirect traffic anywhere specific, the attacker floods the network with routing updates to force constant recalculation, degrading or breaking routing entirely.
- Vulnerability exploited: routers must process every routing update they receive from a trusted neighbor. Without authentication, an attacker can generate a high volume of updates cheaply.
- Technique: repeatedly advertising and withdrawing routes, or flooding a high volume of route changes, keeps affected routers permanently busy recalculating instead of forwarding traffic.
- Impact: a routing-layer denial of service, the same underlying goal as Week 4's STP DoS attack, just aimed at the routing protocol instead of the spanning tree.

Routing DoS / Instability Attack: Mitigation

- Route authentication remains the primary fix: an unauthenticated flood of updates is simply rejected before it can force any recalculation at all.
- Route dampening (suppressing a route that flaps repeatedly, so it's temporarily ignored instead of triggering fresh recalculation every time) is a standard operational defense on protocols like BGP.
- Rate-limiting how frequently routing updates are processed provides a backstop even when authentication is somehow bypassed or misconfigured.

Which Attacks Apply to Which Protocols

Attack	RIP	OSPF	EIGRP	BGP
Routing Table Poisoning	False Response, low hop count	Rogue LSA flooded domain-wide	False Update, attractive metric	False UPDATE, favourable attributes
Unauthorised Route Injection	No adjacency needed, just send	Hello-based adjacency, then flood	Hello-based adjacency, then send	Must spoof a configured neighbor
Route/Prefix Hijacking	Local scope only	Local scope only	Local scope only	Primary case: UPDATE claims a prefix
Blackhole / Sinkhole	Via poisoning above	Via poisoning above	Via poisoning above	Via hijacking above
Routing DoS / Instability	Response flood, repeated recalc	LSA flood/flap, repeated SPF runs	Query flood, repeated DUAL runs	Route flapping, global recalc

Prefix hijacking is the one attack that's genuinely BGP-specific; the other four have an analog on every protocol, just via a different exact mechanism.

Defenses in Detail

The mechanisms behind this week's mitigations

A Note on Scope

- From here, implementation detail focuses specifically on two protocols: OSPF and EIGRP.
- This isn't an arbitrary choice. These are the two protocols with confirmed, reliable MD5 authentication support in Packet Tracer, which is exactly what you'll configure hands-on in this week's lab.
- RIPv2's authentication support is inconsistent across Packet Tracer versions, and BGP's overall protocol support in Packet Tracer is limited enough that we won't rely on it. Both remain relevant in theory, but neither is something we'll configure and verify this week.
- Where a defense mechanism behaves differently between OSPF and EIGRP specifically, the following slides call that out explicitly, since spotting that kind of difference is exactly the LO2 evaluation skill this module is building.

Neighbor Authentication (MD5)

- The core defense across almost every attack this week: routers only form a neighbor relationship, and only accept routing updates, from a device that can prove it holds the correct shared key.
- This is exactly what you'll configure hands-on in this week's lab, on both OSPF and EIGRP.

Authentication: OSPF vs EIGRP Implementation

- OSPF: authentication is enabled per area (area 0 authentication message-digest) or per interface, with a single MD5 key configured directly on that interface. Straightforward, but only one active key at a time.
- EIGRP: authentication requires defining a key chain first (a named object holding one or more keys), then applying both the authentication mode and the key chain to an interface. More setup steps, but key chains support multiple keys with defined lifetimes, enabling scheduled key rotation without downtime.
- The LO2 takeaway: EIGRP's extra configuration complexity buys a real operational capability (key rotation) that OSPF's simpler mechanism doesn't offer out of the box. That's a genuine cost/capability trade-off, not just implementation trivia.

Passive-Interface

- Stops a router from sending (and in most implementations, also processing received) routing protocol traffic out a specific interface, while that interface's network is still advertised to the rest of the routing domain.
- Ideal for interfaces facing end users or other non-router devices. There's no legitimate reason for a routing update to arrive from a printer or a PC.
- A simple, low-cost control that directly removes the attack surface on interfaces that never needed to participate in routing in the first place.
- OSPF and EIGRP implementation is essentially identical here: the passive-interface command sits under the routing process and names the interface, in both cases. This is one control where there's no meaningful difference to evaluate between the two.

Route Filtering / Distribute-Lists

- Explicitly defines which specific routes are acceptable to learn from, or advertise to, a given neighbor, rather than trusting everything a neighbor sends.
- Directly stops implausible or out-of-scope advertisements even from an authenticated neighbor. Authentication proves who's talking; filtering controls what they're allowed to say.
- This is exactly the control PCCW should have had in place in the 2008 YouTube hijack: filtering Pakistan Telecom's advertisements to only the prefixes they were actually authorised to announce.

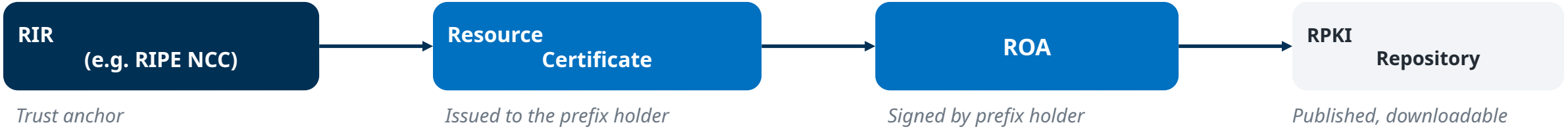
Route Filtering: A Protocol-Type Caveat

- EIGRP: because it's distance-vector, a distribute-list directly controls what actually goes into the Update messages sent to, or accepted from, a neighbor. Filtering here genuinely stops a route from propagating further.
- OSPF: because it's link-state, every router in an area needs an identical copy of the link-state database to calculate correct paths. A distribute-list on OSPF only filters what gets installed into the local routing table, not what's flooded on to other routers via LSAs.
- The LO2 takeaway: the same control name (‘distribute-list’) does a meaningfully different job depending on the protocol's type. On OSPF, actually stopping propagation needs a different tool (e.g. an area filter-list at an area boundary), not a plain distribute-list.

RPKI (Resource Public Key Infrastructure)

- A cryptographic system letting the legitimate holder of an IP address block publish a signed Route Origin Authorisation (ROA), stating which AS is allowed to originate routes for it.
- Other networks can then validate a BGP advertisement against the ROA before accepting it, directly closing the gap that made the YouTube hijack possible.
- BGP-specific, and adopted unevenly across the real Internet even today. Theory-only for this module, since it isn't something Packet Tracer can demonstrate.

RPKI: How the Trust Chain Works



- RIRs (Regional Internet Registries, e.g. RIPE NCC, ARIN, APNIC) act as trust anchors. Each one issues an X.509 resource certificate that cryptographically binds an IP prefix to whichever organisation was legitimately allocated it.
- The prefix holder then creates and signs a Route Origin Authorisation (ROA): a small, specific statement declaring exactly which AS number is authorised to originate routes for that prefix, and the maximum prefix length allowed.
- ROAs are published into a distributed set of RPKI repositories that any network operator can download and cryptographically verify, using the RIR's certificate as the root of trust.
- This is genuinely different from just trusting a BGP announcement: a ROA is a signed, independently verifiable statement made in advance, not something an attacker can fake without the real prefix holder's private key.

RPKI: Route Origin Validation in Action

- A router (or dedicated validator software feeding it) checks every incoming BGP announcement against its local cache of validated ROAs, producing one of three outcomes.

Valid	Invalid	NotFound
Announced AS and prefix length both match a ROA	A ROA exists, but the AS or length doesn't match it	No ROA exists for this prefix at all
Accepted normally	*Rejected. This is exactly what a hijack looks like*	*Treated as normal, unvalidated routing*

- Applied to 2008: had PCCW validated Pakistan Telecom's announcement of 208.65.153.0/24 against a ROA showing AS36561 as the only authorised origin, the announcement would have come back Invalid and simply been dropped, never reaching the rest of the Internet.
- Two honest limitations: RPKI only validates the origin AS, not the full AS-path, so it doesn't stop every possible manipulation. And it only works where operators have both registered ROAs and actually enabled validation, adoption remains uneven across the real Internet even today.

Attack Summary: Vulnerability, Exploit, Impact, and Mitigation

Attack	Vulnerability	Exploit	Impact(s)	Mitigations
Routing Table Poisoning	No verification of neighbor route claims	False advertisement with an attractive (low) metric	Misrouted traffic; denial of service; interception	MD5 authentication; route filtering; passive-interface
Unauthorised Route Injection	No authentication required to form a neighbor adjacency	Rogue device joins as a full, trusted neighbor	Persistent foothold; topology visibility; ongoing false routes	MD5 authentication; passive-interface; physical port security
Route/Prefix Hijacking	No verification of prefix ownership (BGP-specific)	Advertise an unowned, often more specific, prefix	Global traffic redirection; denial of service; interception	RPKI; provider-level route filtering
Blackhole / Sinkhole	Same lack of route verification as poisoning/injection	Redirect traffic via manipulated routing information	Denial of service (blackhole); interception and tampering (sinkhole)	MD5 authentication; filtering; passive-interface; traffic monitoring
Routing DoS / Instability	No authentication; must process every received update	Flood routing updates, or repeatedly advertise and withdraw routes	Router CPU exhaustion; network-wide DoS; unstable forwarding	MD5 authentication; route dampening; rate-limiting

Full Mitigation Comparison

Attack	Primary mitigation
Routing Table Poisoning	Neighbor authentication (MD5), route filtering
Unauthorised Route Injection	Neighbor authentication (MD5), passive-interface
Route/Prefix Hijacking	RPKI, provider-level route filtering (theory only)
Blackhole / Sinkhole Attack	Same as injection/poisoning, plus traffic monitoring
Routing DoS / Instability Attack	Neighbor authentication (MD5), route dampening, rate-limiting

Layered Defense for Routing

- Notice how much weight neighbor authentication alone carries here: it's the single control that meaningfully addresses four of the five attacks covered today.
- But authentication alone doesn't stop everything: it proves who's talking, not that what they're saying is plausible. That's what filtering adds on top.
- Same layered-defense principle from every week so far: authentication, filtering, and passive-interface each close a different gap, and a real network needs more than one of them.

Summary

- Routing protocols automate what would otherwise be manual configuration, but that automation runs entirely on trust between neighbors.
- RIP, OSPF, EIGRP, and BGP each build that trust differently, but none of them verify it by default.
- Five attacks (poisoning, unauthorised injection, hijacking, blackhole/sinkhole, and routing DoS) all trace back to that same unverified trust, aimed at different outcomes.
- The 2008 YouTube hijack shows this isn't hypothetical: a local censorship attempt with no malicious global intent still took down a major service worldwide for two hours.
- Neighbor authentication (MD5) is this week's strongest single control, and the one you'll actually configure on OSPF and EIGRP in the lab.

This Week's Lab & Tutorial

- Lab (runs in the Week 6 session): configure and verify MD5 neighbor authentication on both OSPF and EIGRP in Packet Tracer, the two protocols confirmed to support this reliably.
- Tutorial worksheet: study-notes-style questions covering the four protocols, all five attacks, and their mitigations. Use it to prep for the mid-semester test.

References

- Santos & Stuppi, CCNA Security 210-260 Official Cert Guide, 1st Ed., Cisco Press.
- Kaufman, Perlman, Speciner & Perlner, Network Security: Private Communication in a Public World, 3rd Ed., Prentice Hall.
- RIPE NCC, ‘YouTube Hijacking: A RIPE NCC RIS Case Study’, 2008.
- Cisco Networking Academy, CCNA Security curriculum (routing protocol security chapters).